

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative AI Engineer with 8+ years of professional experience in delivering machine learning, natural language processing, and end-to-end AI solutions, from R&D to production deployment. Specializes in text-based models, unstructured data pipelines, and compliance-aware systems, particularly within highly regulated sectors. Proven record of shipping scalable, production-grade ML applications that integrate Python, AWS, and modern MLOps practices. Fluent in English with strong stakeholder communication skills and a business analyst mindset.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (Pandas, NumPy, Scikit-learn, NLTK, SpaCy, PyTorch, TensorFlow), SQL, JavaScript, TypeScript, SCSS, Bash
- **Machine Learning & NLP:** Named Entity Recognition (NER), Text Classification, Topic Modeling, Document Summarization, Transformers (HuggingFace)
- **Cloud & DevOps:** AWS (S3, Lambda, SageMaker, EC2), GCP (Vertex AI, BigQuery), Docker, Kubernetes, GitHub Actions, Airflow, MLflow
- **Data & Storage:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch
- **Tools & Platforms:** Jupyter, VSCode, Apache Spark, FastAPI, Flask, TensorBoard, Label Studio
- **Compliance & Security:** GDPR, HIPAA-aligned practices, data versioning, model auditing
- **Other:** CI/CD, A/B testing, microservices, Agile/Scrum

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Delivered multi-stage NLP pipelines processing pharmaceutical and clinical documentation, implementing NER and summarization aligned with industry-specific compliance.
- Designed and deployed scalable Python-based ML services with model versioning, monitoring, and rollback strategies for real-world production environments.
- Led initiatives transforming R&D prototypes into reliable production models by aligning stakeholders across legal, data science, and business teams.
- Built automated MLOps workflows using MLflow and SageMaker, improving reproducibility and reducing deployment times by 40%.
- Conducted large-scale data wrangling and annotation strategies for unstructured biomedical datasets using Spark and custom labeling interfaces.

- Collaborated with domain experts to ensure alignment with trial protocols, stability analysis, and safety reporting standards in pharma environments.
- Architected end-to-end pipelines with GCP and AWS, integrating clinical NLP models with API services and real-time inference layers.
- Instituted monitoring dashboards for drift detection, model performance, and compliance auditing to meet regulatory standards.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Created custom NLP pipelines that extracted structured data from trial documentation, safety forms, and regulatory text using SpaCy and HuggingFace transformers.
- Designed and implemented privacy-preserving text processing tools to redact PII from sensitive patient and clinical notes using AWS Lambda and Comprehend.
- Developed document search systems combining Elasticsearch and sentence embeddings to retrieve pharma-related content with 85%+ accuracy.
- Collaborated closely with business analysts to refine AI use cases based on customer feedback, legal reviews, and operational KPIs.
- Handled version control, validation sets, and production shadow testing to meet stringent healthcare model governance policies.
- Supported on-prem and cloud hybrid setups for pharmaceutical clients requiring data locality and air-gapped inference environments.
- Developed RESTful APIs with FastAPI for document classification, summarization, and risk flagging.
- Built explainable AI components for models used in regulatory approval processes, increasing trust and adoption by clinical teams.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Built analytics dashboards and backend APIs in Node.js and Python for logistics and healthcare products.
- Created document parsing engines using regex, OCR, and NLP to extract structured metadata from pharma forms.
- Integrated NoSQL and relational databases for managing patient surveys, lab forms, and physician notes.
- Implemented multi-language support and accessibility features (WCAG) in UI components for medical tools.
- Designed audit logging and error recovery systems for cloud-native workflows deployed in AWS.
- Implemented access control and JWT-based authentication for regulated data access.
- Optimized data pipelines for mobile clinical apps reducing API latency by 45%.
- Collaborated on internal compliance workshops to ensure developers understood ISO and HIPAA standards.

Software Developer

Decima Digital

November 2015 – April 2016

Tallinn, Estonia

- Contributed to early-stage healthcare application focused on scheduling and managing pharmaceutical visits.
- Implemented backend APIs in Flask and integrated with frontend Angular components.
- Handled system logs and access security for deployments on EC2 Linux instances.
- Built document ingestion tools for digitized protocols using NLP techniques.
- Collaborated with a QA team to define data validation checks for structured records.
- Configured MongoDB replica sets to ensure high availability of regulated datasets.
- Coordinated user testing sessions with medical professionals for feedback loops.
- Managed API schema updates in Swagger to support third-party integrations.

Software Developer Intern

Borna Technology

September 2014 – August 2015

Tallinn, Estonia (On-Site)

- Designed internal tools to track progress of clinical trial documentation submissions.
- Created scripts to classify scientific abstracts using early ML models.
- Built internal dashboards using Bootstrap and Angular for QA monitoring.
- Automated document parsing of trial protocols using Python and regex libraries.
- Ensured CSV-to-JSON transformation for structured ingestion of clinical notes.
- Shadowed senior developers during regulatory reviews to understand compliance needs.
- Helped refactor backend scripts to transition from legacy PHP to Python.
- Learned early Agile processes and participated in sprint grooming and retrospectives.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Machine Learning – Specialty
- ❖ TensorFlow Developer Certificate
- ❖ Google Cloud Professional Data Engineer
- ❖ Data Science Specialization (Johns Hopkins University, Coursera)

PROJECTS

- ✓ **AI-Powered Regulatory Analyzer**
Built a production-grade NLP pipeline to classify, summarize, and extract key data points from

regulatory and clinical trial submissions. Integrated with AWS SageMaker for scalable inference, and passed internal pharma compliance audits.

✓ **Pharma Document Search Engine**

Developed a semantic search engine for unstructured pharmaceutical data using Elasticsearch, BERT embeddings, and streamlit interface for internal analysts. Supported multiple filters, drug aliases, and trial phases.

✓ **Sensitive Data Scrubber**

Created a data preprocessing tool to detect and mask sensitive information (PHI/PII) in text streams. Leveraged named entity recognition (spaCy + custom rules) and conformed to GDPR and HIPAA masking rules.